

Exp: 08-Write the Python Program to Implement DFS.

Aim:

To write a python program to implement DFS.

Algorithm:

Step 1: Initialize a set to keep track of visited nodes.

Step 2: Call the DFS-Visit function with the starting node and the set of visited nodes.

Step 3: In the DFS-Visit function, add the current node to the set of visited nodes.

Step 4: For each neighbor of the current node, if it has not been visited yet, recursively call the DFS-Visit function with the neighbor node and the set of visited nodes.

Step 5: Repeat steps 3-4 until all reachable nodes have been visited.

Program:

```
graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': ['F'],
    'F': []
}

def dfs(graph, node, visited):

    if node not in visited:

        print(node, end=" ")

        visited.add(node)

        for neighbor in graph[node]:
            dfs(graph, neighbor, visited)

visited = set()

print("DFS Traversal:")
```

```
dfs(graph, 'A', visited)
```

Output:

```
DFS Traversal:
A B D E F C
```

Result:

Thus, the Program was executed and output found to be correct.